# FIITJEE
# ALL INDIA TEST SERIES
## JEE (Advanced)-2023
## PART TEST – II
## PAPER –2
## TEST DATE: 11-12-2022

**Time Allotted: 3 Hours** **Maximum Marks: 180**

**General Instructions:**

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

**Section – A *(01 – 04, 19 – 22, 37 – 40)*:** This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.

**Section – A *(05 –10, 23 – 28, 41 – 46)*:** This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

**Section – B *(11 – 18, 29 – 36, 47 – 54)*:** This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.**

**MARKING SCHEME**

**Section – A (Single Correct):** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| *Full Marks* | : | +3 | If ONLY the correct option is chosen. |
| Zero Marks | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| *Negative Marks* | : | –1 | In all other cases. |

**Section – A (One or More than One Correct):** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| Full Marks | : | +4 | If only (all) the correct option(s) is (are) chosen; |
| Partial Marks | : | +3 | If all the four options are correct but ONLY three options are chosen; |
| Partial marks | : | +2 | if three or more options are correct but ONLY two options are chosen and both of which are correct; |
| Partial Marks | : | +1 | If two or more options are correct but ONLY one option is chosen and it is a correct option; |
| Zero Marks | : | 0 | If none of the options is chosen (i.e. the question is unanswered); |
| Negative Marks | : | –2 | In all other cases. |

**Section – B:** Answer to each question will be evaluated according to the following marking scheme:

| | | | |
|---|---|---|---|
| Full Marks | : | +3 | If ONLY the correct integer is entered; |
| Zero Marks | : | 0 | Question is unanswered; |
| Negative Marks | : | –1 | In all other cases. |

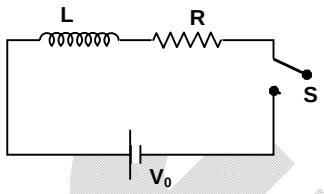
# Physics

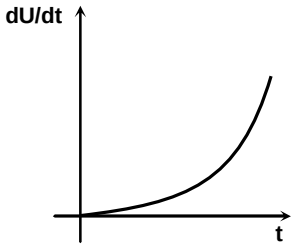
## PART – I

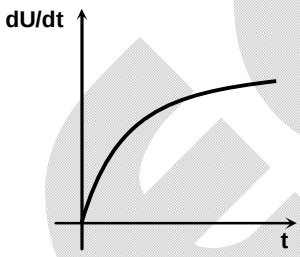
### Section – A (Maximum Marks: 12)

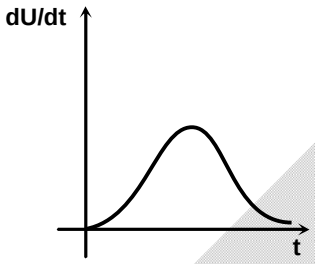
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

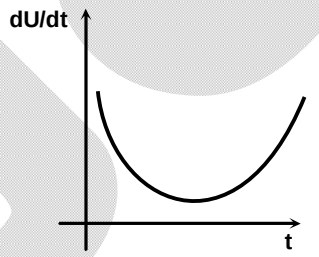
1. An ideal inductor and a resistor are connected in series with a battery of pontential difference $V_0$ through a switch 'S'. The switch is closed at t = 0. The potential energy stored in inductor at time 't' is given by U. The rate of change of potential energy in inductor with time is best represented by



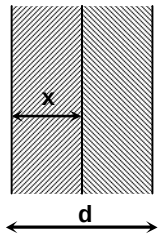

(A)


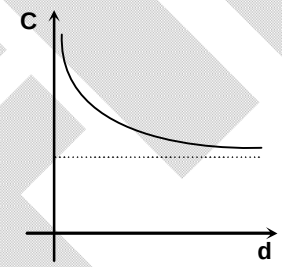

(B)


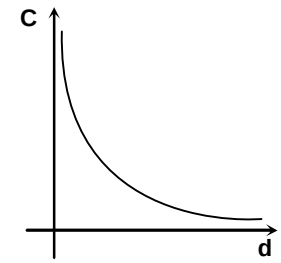

(C)


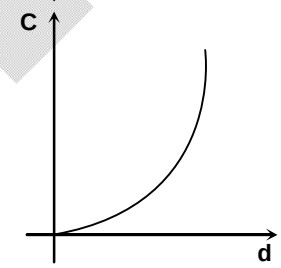

(D)


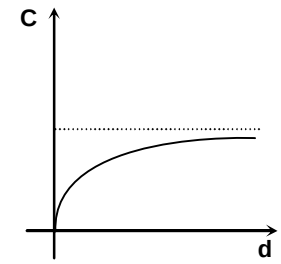

2. A dielectric is completely filled between the space of a parallel plate capacitor. The separation between plates of capacitor is 'd' and dielectric constant is given by $k = k_0 e^{\lambda x}$, where '$k_0$' and '$\lambda$' are positive constants and x is distance from one plate of capacitor. If distance between plate (d) increases slowly then graph between capacitance of capacitor and distance between plate is given by



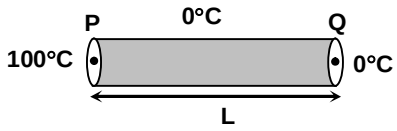

(A)


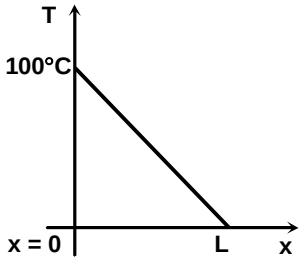

(B)


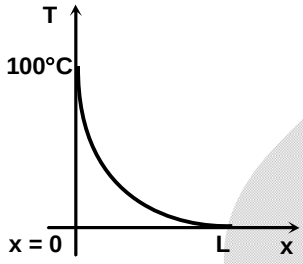

(C)


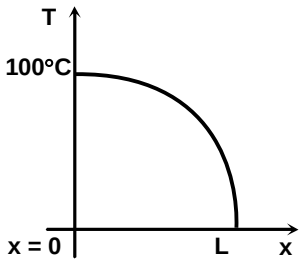

(D)


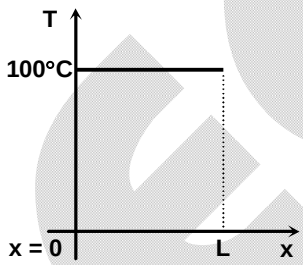
3. A conducting rod of length 'L' cross-sectional area 'A' and thermal conductivity 'k' is kept in the surrounding at 0°C. The end P is maintained at 100°C and Q at 0°C. The graph between temperature (T) of the rod and distance (x) from the end 'P' is best represented by



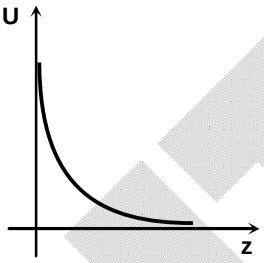

(A)


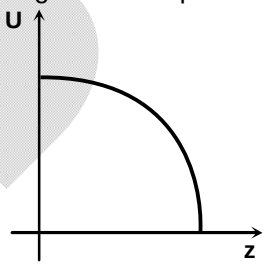

(B)


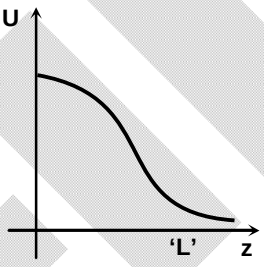

(C)


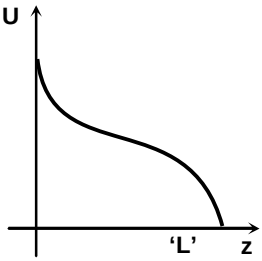

(D)



4. Four point charges 2q, 2q, –q and –q are kept at (a, –a, 0), (–a, +a, 0), (a, a, 0) and (–a, –a, 0) and fixed at their position. Another point charge $q_0$ is kept at origin and slightly displaced along positive z-axis. The graph of potential energy of the charge is best represented by

(A)



(B)


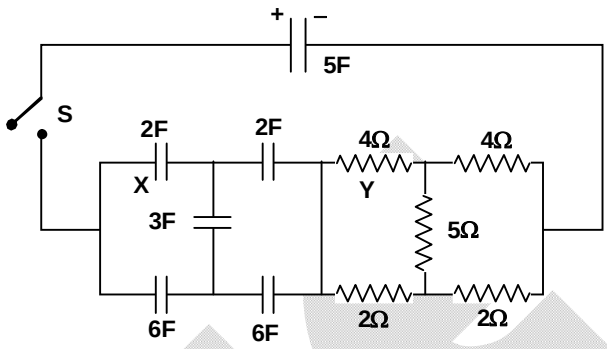

(C)



(D)


## Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

5. A capacitor of capacity 5F is charged by a battery of 2 volt and then connected with 5 uncharged capacitors and 5 resistors with the help of a switch 'S' as shown in the figure. At t = 0, switch is closed. Choose the correct option(s).



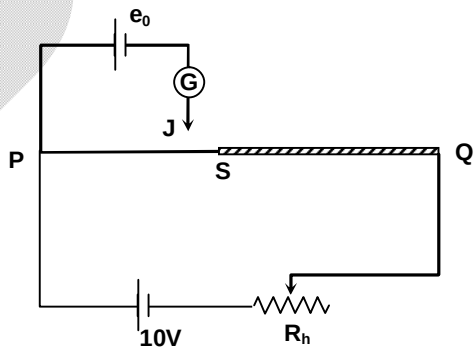

(A) Charge on capacitor X at time t is given by $\frac{10}{9}\left(1-e^{-\frac{27}{160}t}\right)$

(B) Current in resistor Y at time t is given by $\frac{1}{4}\left(e^{-\frac{27}{160}t}\right)$

(C) Total energy dissipated in the circuit is $\frac{50}{9}$J

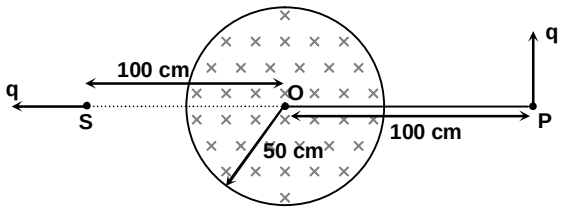
(D) Total energy dissipated in the circuit is $\frac{40}{9}$J

6. A potentiometer wire PQ is made by using two wires of different cross sectional area A and 2A. The material of wire PS and SQ is same and their lengths are 5m each. A source battery of 10V and rheostat $R_h$ are connected in the circuit as shown in the figure. Choose the correct option(s) regarding the balance point.




(A) If a cell of emf $e_0$ = 5V is used then balance point from the end P is at 7.5 m
(B) If a cell of emf $e_0$ = 5V is used then balance point from the end P is at 3.75 m
(C) If a cell of emf $e_0$ = 8V is used then balance point from the end P is at 7 m
(D) If a cell of emf $e_0$ = 8V is used then balance point from the end P is at 8.5 m

7. A cylindrical vertical magnetic field exist in a gravity free space. Magnetic field is increasing at a rate of 700 T/sec. The radius of cylindrical magnetic field is 50 cm. A point charge q = 50 μC is moved very slowly by an external agent from two point 'P' and 'S' then

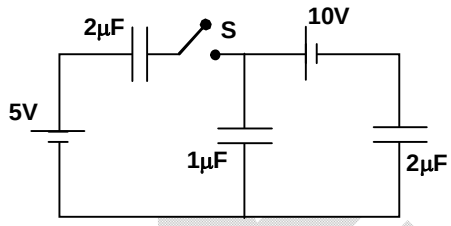



(A) Work done by external agent is zero if charge particle is moved from point 'S' upto infinity along the line OS.
(B) Work done by external agent is zero if charge particle is moved from point 'P' upto infinity perpendicular to the line OP.

(C) Work done by external agent is $\frac{55}{8}$mJ if charge particle is moved from point 'P' upto infinity perpendicular to the line OP.

(D) Work done by external agent to move a charge particle from any point to infinity is path independent.

8. Three capacitors of capacity 2 μF, 2 μF, 1μF and two ideal cells of potential difference 10V and 5V are connected in a circuit through a switch 'S'. Initially switch is opened for a long time and then it is closed. Choose the correct option(s) after switch is closed.



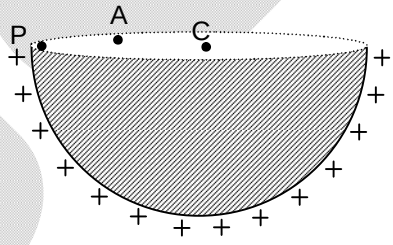

(A) Charge passing through 10V cell is $\frac{4}{3}\mu C$

(B) Work done by 5V battery is 10μJ

(C) Charge on 1μF capacitor is 6μC

(D) Total heat dissipated in the circuit after switching is $\frac{5}{3}\mu J$

9. A non conducting hemispherical shell of radius 'R' has uniform surface charge density σ. Point 'C' is center of hemispherical shell. Point 'A' and point 'P' lie on the flat surface of the shell. The distance between C and A is R/2 and between C and P is R. Choose the correct option(s).



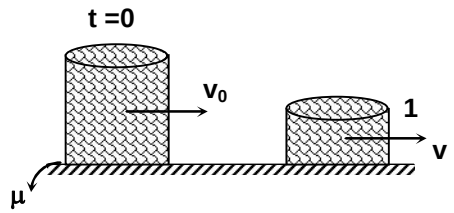

(A) The potential at point P is $\frac{\sigma R}{2\varepsilon_0}$

(B) The potential at point A is $\frac{\sigma R}{4\varepsilon_0}$

(C) Word done by electrostatic force if a charge '$q_0$' is moved from P to A is $\frac{-q_0\sigma R}{4\varepsilon_0}$

(D) Word done by electrostatic force if a charge '$q_0$' is moved from P to A is $\frac{q_0\sigma R}{4\varepsilon_0}$

10. An ice in the shape of cylinder is kept on a rough horizontal surface as shown in the figure. An initial velocity $v_0$ is imparted to cylindrical ice. The initial mass of ice is $m_0$ and coefficient of friction between ice and horizontal surface is 'μ'. The initial temperature of ice 0°C and its latent heat of fusion is L. Choose the correct option(s).




(A) The velocity of cylindrical ice at time 't' is given by $(v_0 + \mu gt)$

(B) The velocity of cylindrical ice at time 't' is given by $(v_0 - \mu gt)$

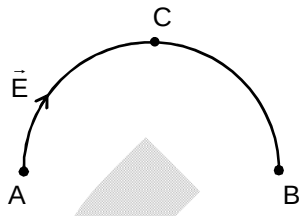
(C) The mass of cylindrical ice at time 't' is given by $m_0 e^{-\frac{\mu g}{L}\left(v_0 t - \frac{\mu g L^2}{2}\right)}$

(D) The mass of cylindrical ice at time 't' is given by $m_0 e^{-\frac{\mu g}{L}\left(\frac{v_0 t}{2} - \frac{\mu g L^2}{2}\right)}$

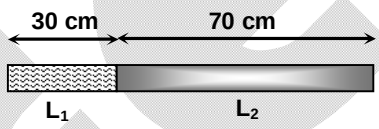
**Section – B (Maximum Marks: 24)**

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

11. Semicircular electric field line is present in a space. The magnitude of electric field is constant at each point of electric field lines. A point charge is constrained to move along the filed lines. The particle starts from rest and initially it is at point A. The magnitude of acceleration of the particle at C is $a_1$ and it is $a_2$ when reaches at B. The ratio of $a_1$ and $a_2$ is given by $\sqrt{\frac{1+k_1}{1+k_2}}$. Find the value of $\frac{k_2}{k_1}$.

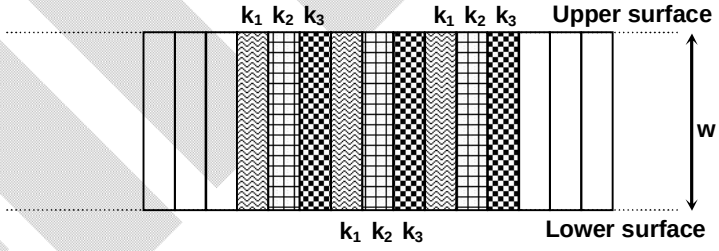



12. Two rods $L_1$ and $L_2$ of different materials are used to make a single composite rod as shown in the figure. The cross sectional are of both rods is same. At 25°C, the total length of combined rod is 1m, in which the length of $L_1$ is 30 cm. When temperature of combined rod is increased, at 125°C, the total length of combined rod increased by 1.91 mm. The coefficient of linear expansion of $L_1$ is $1.7\times10^{-5}$/°C and that of the $L_2$ is $k\times10^{-5}$/°C. Find the value of k.

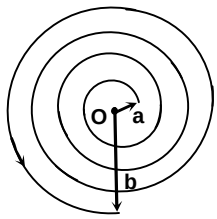



13. A particle of mass m = $\sqrt{227}$ gm and charge q = 7μC is projected in a space where magnetic field is given by $\vec{B} = y^2\hat{i} + z^2\hat{j} + x^2\hat{k}$. The initial velocity of projection of particle is $\vec{v} = \hat{i} - 2\hat{j} + 3\hat{k}$ from (1, 2, 1). Find radius of curvature of particle at t = 0 in kilometer.

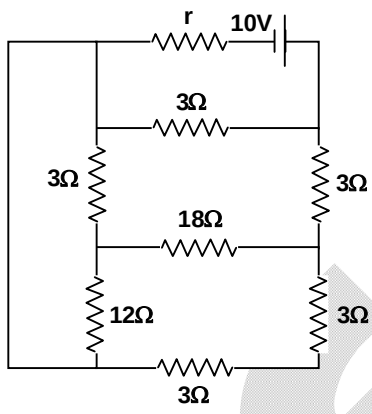
14. Three different type of glasses are used to make a large glass slab. The thermal conductivity of glass slabs are $k_1$, $k_2$ and $k_3$. The width of each glass slab is 'ω' and cross-sectional area is A. The glass slabs are kept in a pattern as shown in the figure. The upper surface of glass slab is maintained at temperature $T_1$ and lower surface at $T_2$. It is given that $k_1$ = 1.10 W/mK, $k_2$ = 2.82 W/mK, $k_3$ = 2.08 W/mK, w = 10 cm, A = 1.3 $m^2$, $T_1$ = 100°C, $T_2$ = 0°C. Find equivalent thermal conductivity of large glass slab in W/mK.



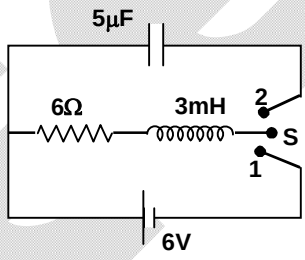

15. A conducting spiral of inner radius a = 20 cm and outer radius b = 40 cm is kept on a frictionless horizontal surface. The total number of turns is equal to 1000 which are wound tightly and insulated to each other. There exist a vertical magnetic field which is changing at a rate of $\frac{50}{\pi}$ mT/sec. If the resistance of conducting spiral is $\frac{14}{9}\Omega$, then find induced current in ampere.



16. A battery of potential difference 10 V and internal resistance 'r' is connected to seven resistors as shown in the figure. If battery delivers maximum power to the circuit with internal resistance $r = \frac{4}{13}\lambda$, then find the value of $\lambda$.

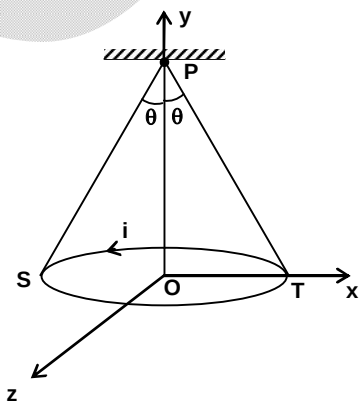



17. A capacitor (5μF), a resistor (6Ω), an inductor (3mH), a cell (6V) and a switch are connected in a circuit as shown in the figure. Initially switch is kept at position-1 for a long time. The potential difference (in volt) across the ends of inductor immediately after the switch is shifted from position-1 to position-2 is




18. A ring of uniform mass density (m = 100 gm) and radius (R = 50 cm) is suspended by two strings PS and PT from the point P on y-axis in vertical plane. The ring is kept parallel to horizontal surface on x-z plane. Ring carries a current $\left(i = \frac{7}{22}\text{amp}\right)$ as shown in the figure. There exist a magnetic field (B = 1 Tesla) along negative x-axis. The tension in the string PS is $T_1$ and in the string PT is $T_2$. Find the ratio $\frac{T_1}{T_2}$.

(It is given that $\theta = 60°$)



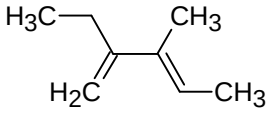
# Chemistry

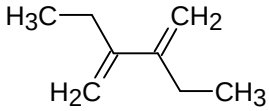
## PART – II

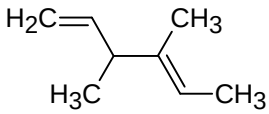
### Section – A (Maximum Marks: 12)

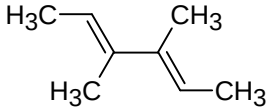
*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

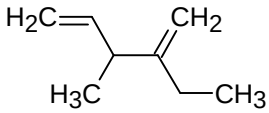
19. Which of the following is NOT the chain isomer of



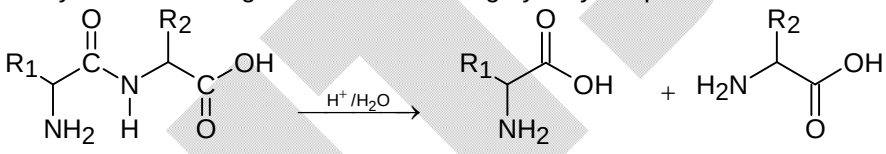
(A) 

(B) 

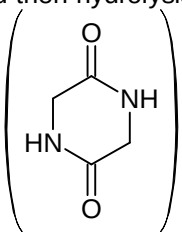
(C) 

(D) 

20. Analyse the following data of the following hydrolyzed products and choose the correct option:



Dipeptide [x] [y]

If the dipeptide is first treated with $HNO_2$ and then hydrolysis is carried out, we got lactic acid and [y]. [y] on heating gives 2,5-diketopiperazine 

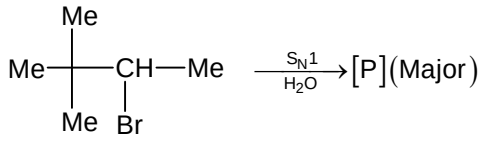
(A) $R_1 = H$ and $R_2 = CH_3$

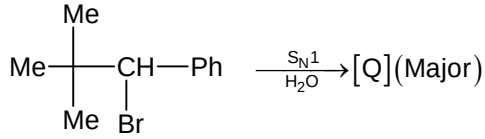
(B) $R_1 = CH_3$ and $R_2 = CH_3$

(C) $R_1 = CH_3$ and $R_2 = H$

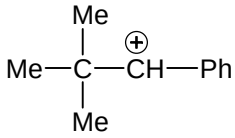
(D) $R_1 = H$ and $R_2 = H$

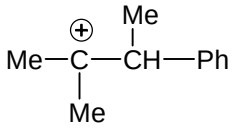
21.

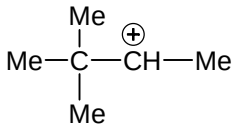
(I) 

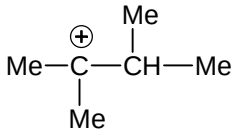
(II) 

Identify the INCORRECT intermediate carbocation, which does NOT form during the reaction (I) and (II)?

(A) 

(B) 

(C) 

(D) 

22. Identify which test are positive (+) or negative (–) for $\beta$ – D – glucopyranose?

[Note: Here positive means respond to test and negative means not respond to test)

1. 2, 4-DNP
2. $NH_2OH / H^+$
3. Tollen's test
4. $NaHSO_3$

(A)

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| + | + | + | – |

(B)

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| + | + | – | – |

(C)

| 1 | 2 | 3 | 4 |
|---|---|---|---|
| – | + | + | – |

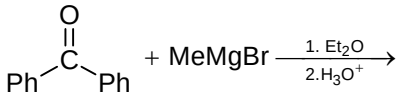
(D)

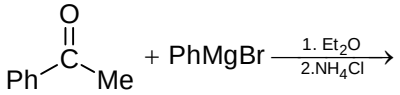
| 1 | 2 | 3 | 4 |
|---|---|---|---|
| – | – | – | – |

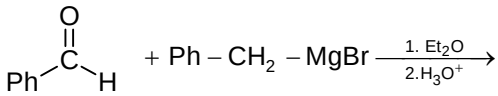
## Section – A (Maximum Marks: 24)

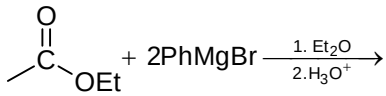
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

23. Which of the following reaction(s) will provide a synthesis of 1,1-diphenylethanol as a major product?

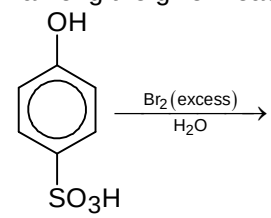
(A) 

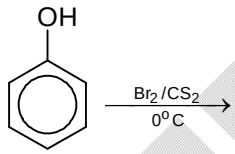

(B) 

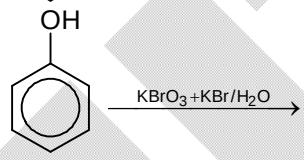

(C) 

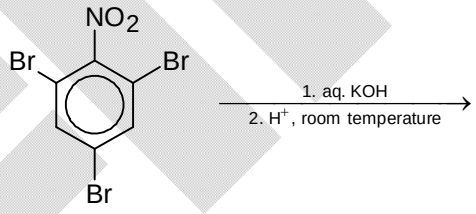

(D) 


24. Which among the given reactions can yield the 2, 4,6-tribromo phenol as a major product?

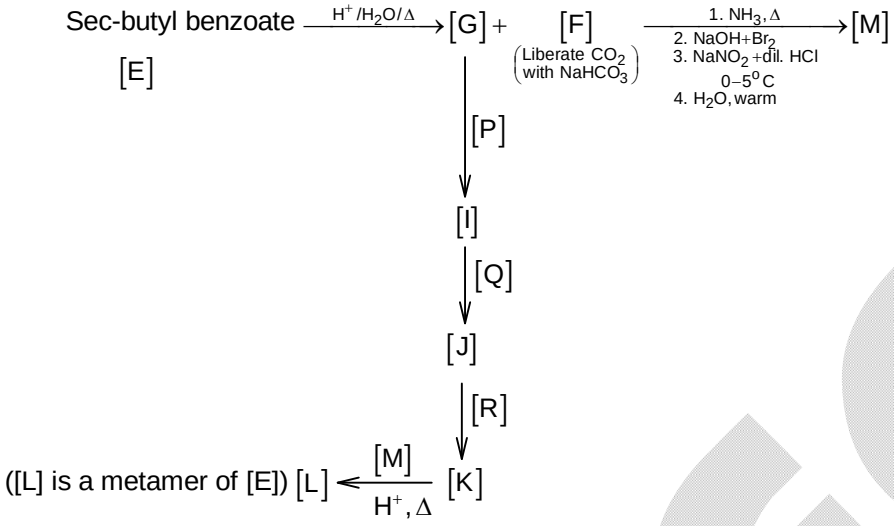
(A) 


(B) 


(C) 


(D) 

25. Choose the correct options regarding the given scheme?




(A) $[P] = SOCl_2$ $[Q] = Mg$ / dry ether $[R] = 1.\ HCHO\ \ 2.\ H^+$

(B) $[P] = KBr + H_2SO_4$ $[Q] = AgCN$ / DMF $[R] = H_3O^+ / \Delta$

(C) $[P] = PBr_3$ $[Q] = Mg$ / dry ether $[R] = 1.\ CO_2\ \ 2.\ H^+$

(D) $[P] = HBr$ $[Q] = NaCN$ / DMF $[R] = H_3O^+ / \Delta$

26. Choose the INCORRECT statement(s) for the given reaction scheme

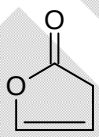
$$\underset{C_4H_4O_2}{[P]} \xrightarrow[2.\ H^+(\text{work up})]{1.\ NaOH} \underset{C_4H_6O_3}{[Q]} \xrightarrow[-CO_2]{\Delta} [R] \text{ gives iodoform test} \xrightarrow{NaOH/\Delta} \underset{C_6H_{10}O}{[S]}$$

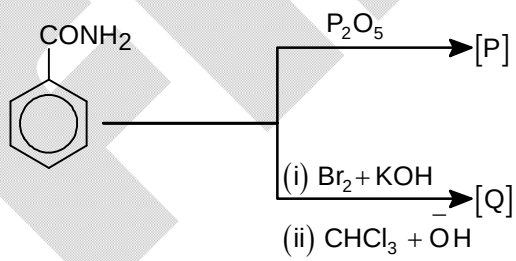
Unsaturated Lactone

(A) [S] is an $\alpha,\beta-$unsaturated aldehyde.

(B) Four stereoisomers are possible for [S].

(C) Two structural isomers of [P] can satisfy the above reaction scheme.

(D) [P] is 

27. 


Which statement(s) is/are CORRECT?

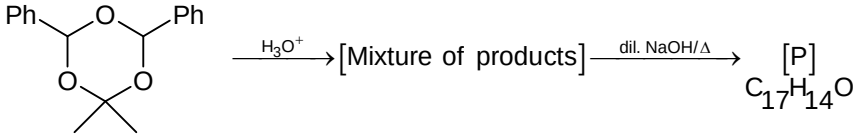
(A) Reduced product of P and Q will be metamers to each other. (Consider $LiAlH_4$ as a reducing agent)

(B) By dry distillation of hydrolysed product of P with $Ca(OH)_2$ gives benzophenone.

(C) One of the hydrolysis product of Q reacts with $NaNO_2$ + HCl followed by reaction with phenol in mild basic medium gives orange red dye.

(D) Electrophile involved in the formation of Q is dichlorocarbene.

28. Choose the correct statement(s)

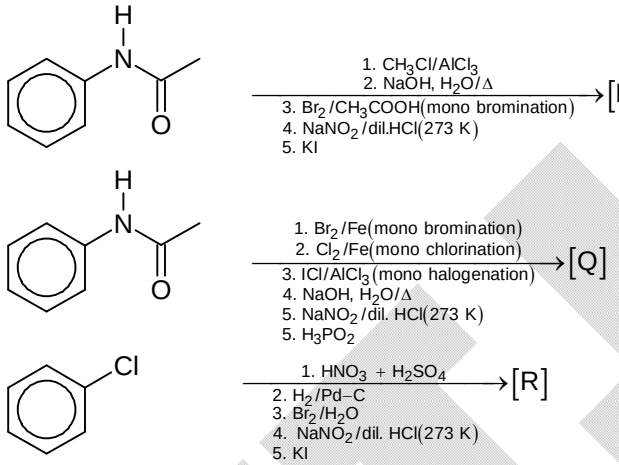



(A) In mixture, one of the product shows positive iodoform test.
(B) In mixture, one of the product shows positive Tollen's test and negative Fehling's test.
(C) [P] on reaction with hydroxyl amine gives oxime that can have 4 stereoisomers.
(D) Number of double bonds in [P] is 10.

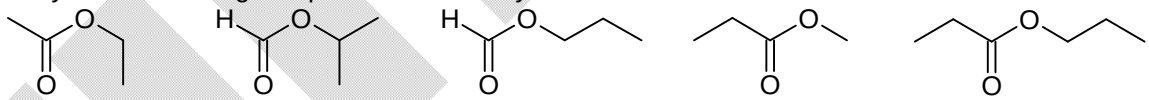
**Section – B (Maximum Marks: 24)**

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

29. Consider the major organic product in each case and calculate the Sum of number of halogen atoms in [P], [Q] and [R]




30. An organic compound reacts with excess of MeMgBr followed by hydrolysis to give two organic products of which one reacts with NaOI to give yellow precipitate and OTHER DOES NOT. How many of the following compounds can satisfy these conditions?




(I)




(II)



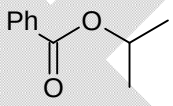

(III)



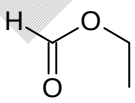

(IV)



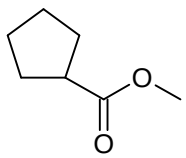

(V)



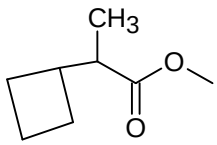

(VI)




(VII)




(VIII)




(IX)

31. Consider all possible isomeric ketones including stereoisomers of MW = 100. All these isomers are independently reacted with $CH_3MgBr$ (Note : Stereoisomers are also reacted separately and consider no racemisation of alpha carbon of the carbonyl group during the reaction). The total number of ketones that gives a racemic product is/are

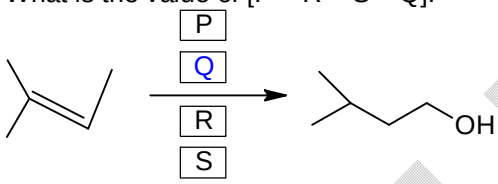
32. $D-Glucose \xrightarrow[H_2O]{Br_2} P$

Number of chiral centre in compound P is

33. Number of correct statement(s) is/are

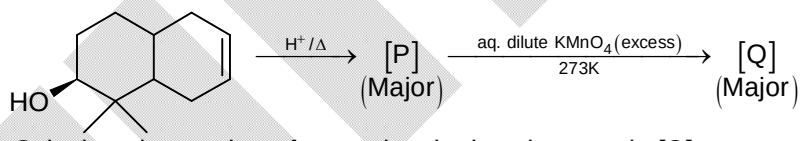
1. Chlorofluorocarbons are responsible for ozone layer depletion.
2. Green house effect is responsible for global warming.
3. Ozone layer does not permit infrared radiation from sun to reach the earth.
4. Acid rain is mostly because of oxides of nitrogen and sulphur.
5. Classical smog occurs in cool humid climate.
6. Classical smog is a mixture of smoke, fog and sulphur dioxide.
7. Classical smog is also called reducing smog.
8. Classical smog is also called oxidizing smog.
9. Ozone absorbs infrared radiations.
10. Ozone absorbs the intense ultraviolet radiations of the sun.

34. Choose the appropriate reagent sequence from the given reagent pool to complete the synthetic conversion. Make sure to use strictly four reagents, not more not less.
Note: You can simply mention the numerical values corresponding to the reagent in the boxes P, Q, R, S respectively and also consider major organic product in each step.
What is the value of [P + R + S – Q].




| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. | HBr | 2. | $H^+ / H_2O$ | 3. | $Hg(OAc)_2 / H_2O$ | 4. | $BH_3.THF$ |
| 5. | $EtONa / EtOH / \Delta$ | | | 6. | $HBr / ROOR / \Delta$ | 7. | conc. $H_2SO_4 / \Delta$ |
| 8. | $NaBH_4$ | | | 9. | $H_2O_2$, aq. NaOH | 10. | $t-BuOK / \Delta$ |

35. [A] is a monomer of natural rubber, on reaction with HBr (1 eq)/$CCl_4$ at 298 K, it forms major product [B].
The number of allylic hydrogen of [A] is x and that of [B] is y, the value of $(y - x)$?

36. 


Calculate the number of secondary hydroxyl groups in [Q]

# Mathematics

## PART – III

### Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*

37. In a triangle ABC let a, b, c denote the length of the sides BC, AC and AB respectively, if $8(b^2 + c^2) = 65a^2$, then the value of $\frac{\cot A}{\cot B + \cot C}$ is equal to

(A) $\frac{57}{16}$ (B) $\frac{53}{8}$

(C) $\frac{57}{8}$ (D) $\frac{59}{16}$

38. Three circles $C_1$, $C_2$, $C_3$ with radii $r_1$, $r_2$, $r_3$ ($r_1 < r_2 < r_3$) respectively are given as $r_1 = 4$ and $r_3 = 9$. They are placed such that $C_2$ lies to the right of $C_1$ and touches it externally, $C_3$ lies to the right of $C_2$ and touches it externally. There exists two straight lines each of which is a direct common tangent simultaneously to all three circles, then $r_2$ is equal to

(A) $\frac{13}{2}$ (B) 6

(C) $\frac{72}{13}$ (D) 7

39. Consider the ellipse $E = 9x^2 + 25y^2 = 225$ with foci S and $S_1$ where S lies on positive x-axis. Let the locus of the image of S with respect to any tangent to the ellipse be the curve C. If the line $ky - 2x + 22 = 0$ touches the curve C, then k is equal to

(A) $\pm 1$ (B) $\pm\sqrt{3}$

(C) $\pm\sqrt{5}$ (D) $\pm 4\sqrt{2}$

40. Let A and B be two distinct points on the parabola $y^2 = 8x$ such that a circle with AB as diameter passes through the vertex of the parabola. Tangents at A and B to the parabola intersect at point C. Another pair of tangents are drawn from point C to the parabola $y^2 = kx$. If angle between these tangents is 90°, then k is equal to

(A) 1 (B) 2

(C) 32 (D) 16

### Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MOER THAN ONE** of these four option(s) is (are) correct answer(s).*

41. Let $f(n) = \sum_{k=1}^{n} \cos\left(\frac{2k\pi}{7}\right)$ and $g(m) = \sum_{k=1}^{m} \cos\left(\frac{2^k\pi}{7}\right)$, then which of the following is/are CORRECT?

(A) f(3) = g(3) (B) 2f(3) = f(6)

(C) 2g(3) = g(6) (D) f(8) = g(8)

42. Consider a circle whose radius is 2 and centre is origin. Tangents are drawn from a point $P(\alpha, 2)$ which touches the circle at $A(a_1, a_2)$ and $B(b_1, b_2)$. Another pair of tangents are drawn from the point $Q(\beta, -2)$ which touches the circle at $A(a_1, a_2)$ and $C(c_1, c_2)$. Then which of the following is/are CORRECT? ($\alpha, \beta > 0$)

(A) $b_1 + b_2 + c_1 + c_2 = 0$ (B) $\alpha\beta = 4$

(C) If $\alpha = 4$, then $4a_2 + 3a_1 = 0$ (D) If $\alpha = \beta$, then $a_2 = 0$

43. Consider a parabola P : $x^2 = 4ky$ and a circle C : $x^2 + y^2 - 2\alpha x + 6y + 9 = 0$ (where $\alpha > 0$). The line of shortest distance between them cuts the parabola P at the end point of latus rectum in first quadrant, then which of the following is/are TRUE?

(A) If k = 1, then $\alpha = 6$

(B) If k = 1, then $\alpha = -6$

(C) If k = 2, then $\alpha = 12$

(D) If $\alpha = 20$, then $k = \frac{17}{3}$

44. In a $\Delta PQR$, let p, q, r denote the length of the sides opposite to angles P, Q and R respectively, let s be the semiperimeter. If $s - p = \frac{s-q}{4} = \frac{s-r}{6}$ and area of the circumcircle of the triangle PQR is $\frac{175}{4\sqrt{66}}$, then which of the following is/are CORRECT?

(A) Area of triangle PQR is $4\sqrt{66}$

(B) Radius of incircle of triangle PQR is $2\sqrt{\frac{6}{11}}$

(C) $\sin\frac{P}{2}\sin\frac{Q}{2}\sin\frac{R}{2} = \frac{12}{175}$

(D) Radius of the excircle opposite to vertex P is $\frac{\sqrt{66}}{5}$

45. Define the collections $\{E_1, E_2, E_3, .....\}$ of ellipse and $\{R_1, R_2, R_3, .....\}$ of rectangles as follows:
$E_1 : x^2 + 4y^2 = 32$
$R_1$ : rectangle of largest area with sides parallel to the axes, inscribed in $E_1$
$E_n$ : ellipse $\frac{x^2}{a_n^2} + \frac{y^2}{b_n^2} = 1$ of largest area inscribed in $R_{n-1}$, n > 1
$R_n$ : rectangle of largest area with sides parallel to axes, inscribed in $E_n$, n > 1.
Then which of the following options is/are CORRECT?

(A) $\sum_{n=1}^{N}(\text{area of } R_n) < 64$ for each positive integer N

(B) The ratio of eccentricities of $E_{10}$ and $E_{12}$ is 1 : 4

(C) The length of major axis of $E_6$ is 2

(D) The length of diagonal of $R_6$ is $\frac{\sqrt{5}}{2}$

46. Let the vertex P of $\Delta PQR$ move on the curve xy = 4. If the perpendicular bisectors of side PQ and PR are given by y = 0 and x – y = 0, then which of the following is/are CORRECT?

(A) Locus of centroid of $\Delta PQR$ is $9xy = 18y^2 + 4$

(B) Locus of centroid $\Delta PQR$ is $9xy = 18x^2 + 4$

(C) Locus of orthocentre of $\Delta PQR$ is $xy = 2y^2 + 4$

(D) Circumcentre of $\Delta PQR$ is origin

## Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **Single Digit Integer, ranging from 0 to 9 both inclusive.***

47. $f(\theta) = \sin^3\theta + \frac{1}{3}\sin^3(3\theta) + \frac{1}{9}\sin^3 9\theta + \frac{1}{27}\sin^3 27\theta + .....$ upto 10 terms, then

$4f(\theta) = k\sin\theta - \frac{\sin 3^m\theta}{3^n}$, then k + n – m is

48. Consider the ellipse $E: \frac{x^2}{25}+\frac{y^2}{9}=1$. Tangents are drawn at the ends points of a double ordinate passing through (c, 0) (where 5 < c < 5). The tangents intersect at G. If $\Delta(c)$ = area of $\Delta EFG$, then for $1 \le c \le 3$ if minimum value of $\Delta(c)$ is k, then $\frac{5k}{8}$ is equal to

49. In a triangle PQR, $\angle P = 54°$, $\angle R = 18°$. If S is a point on PR such that area of triangle PSQ is $\left(\sqrt{5}-1\right)$ times the area of triangle RSQ, then ratio of angle $\angle PQS : \angle RQS$ is

50. A circle with centre (6, 12) and of variable radius cuts the hyperbola $x^2 - y^2 = 144$ at the points A, B, C and D. If the locus of the centroid of the $\Delta ABC$ is $(x-\alpha)^2 - (y-\beta)^2 = \gamma^2$, then $|\alpha + \beta - \gamma|$ is equal to

51. Consider the parabola $y^2 = 4x$ and hyperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$. Tangents are drawn from a point P to the parabola $y^2 = 4x$, the chord of contact of these tangents touches the hyperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$. The locus of point P is a conic whose eccentricity is e, then $4e^2$ is equal to

52. An equilateral triangle ABC is inscribed in the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$. If the eccentric angles of the vertices of triangle ABC are $\alpha$, $\beta$, $\gamma$ respectively and the circumcentre of triangle ABC lies on the ellipse $\frac{9x^2}{7a^2}+\frac{9y^2}{7b^2}=1$, then the value of
$\cos\alpha\cos\beta + \cos\beta\cos\gamma + \cos\gamma\cos\alpha + \sin\alpha\sin\beta + \sin\beta\sin\gamma + \sin\gamma\sin\alpha$ is equal to

53. Let the director circle of the hyperbola $H: \frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be denoted by $C_1$ and let director circle of the ellipse $E: \frac{x^2}{a^2}+\frac{y^2}{b^2}=\frac{1}{3}$ be denoted $C_2$. If the circle $C_2$ is also the director circle of $C_1$ and let $e_1$ and $e_2$ be the eccentricities of the ellipse E and hyperbola H, then $\frac{e_2^2}{e_1^2}$ is equal to (a > b)

54. Let a, b, c denote the lengths of the sides opposite to A, B and C in triangle ABC.
If $2a^2 + 4b^2 + 9c^2 = 4ab + 6ac$, then $\cos B = \frac{p}{k}$ (where p and k are coprime), then p – k is equal to